- How much storage would the users need?
- How much computing power would be required to handle their requests?

To avoid supply gaps, we had to order more or faster hardware than needed, causing unnecessary expenses.

On AWS, you can use services on demand. Planning capacity is less and less important. You can scale from one EC2 instance to thousands of EC2 instances. Storage can grow from gigabytes to petabytes. You can scale on demand, thus replacing capacity planning. AWS calls the ability to scale on demand *elasticity*.

Public cloud providers like AWS can offer the needed capacity with a short waiting time. AWS serves more than a million customers, and at that scale, it isn't a problem to provide you with 100 additional virtual machines within minutes if you suddenly need them. This allows you to address another problem: recurring traffic patterns, as shown in figure 17.1. Think about the load on your infrastructure during the day versus at night, on a weekday versus the weekend, or before Christmas versus the rest of year. Wouldn't it be nice if you could add capacity when traffic grows and remove capacity when traffic shrinks? That's what this chapter is all about.

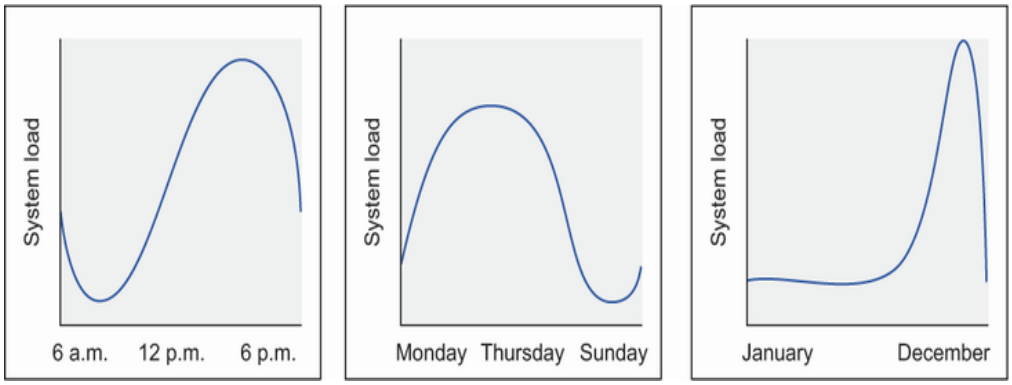


Figure 17.1 Typical traffic patterns for a web shop

Scaling the number of virtual machines is possible with Auto Scaling groups (ASG) and *scaling policies* on AWS. Autoscaling is part of the EC2 service and helps you scale the number of EC2 instances you need to fulfill the current load of your system. We introduced Auto Scaling groups in chapter 13 to ensure that a single virtual machine was running even if an outage of an entire data center occurred.